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Least Squares Methods for Treating Problems with Uncertainty in *x* and *y*

Methods for straight-line fitting of data having uncertainty in *x* and *y* are compared through Monte Carlo simulations and application to specific data sets. Under special circumstances, the "ignorance" methods, methods which are typically used without information about the data errors σ_x and σ_y, are equivalent to the recommended best approach. The latter is numerical rather than formulaic but is easy to implement in programs that permit user-defined fit functions. It can handle any response function, linear or nonlinear, for any σ_{xi} and σ_{yi}. Estimates for the latter must be supplied and rightfully belong in any data analysis.

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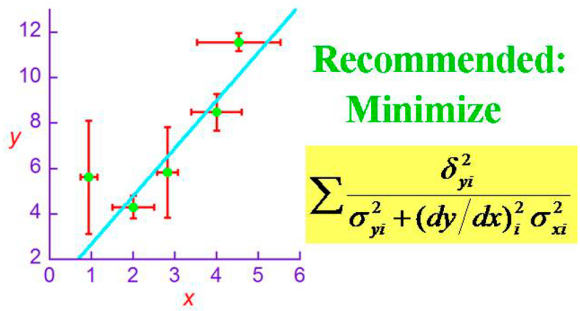

By far the most common use of the method of least-squares (LS) in scientific data analysis is on problems where a single dependent variable (y) is assumed to be uncertain, with one or more independent or predictor variables considered error-free. Then the LS solution is obtained by minimizing the sum[1]

$$S = \sum w_i \delta_{yi}^2 \tag{1}$$

where w_i is the weight and δ_{yi} the residual in y, $\delta_{yi} = Y_i - y_i$, with Y_i the calculated or "adjusted" value of y for the ith values of the independent variables, for which I will henceforth use just x_i. If the relation between the dependent and independent variables is algebraically linear in the adjustable parameters, for example, the 4-parameter model, $f(x) = a + bx + cx^2 + d/x$, the problem is a linear LS one (LLS), solvable in a single computational cycle. If the weights are taken inversely proportional to the data variances, $w_i \propto \sigma_{yi}^{-2}$, the solution is minimum-variance. If further, the random error in y is normally distributed (Gaussian) about the true values y_t of y, the LLS solution is also the maximum likelihood solution. Under these conditions and assuming the fit model is correct, the parameter estimates are normally distributed about their true values (e.g., a_t, b_t), with standard errors (SE) that are given exactly by the covariance matrix $\mathbf{V}_{\text{prior}}$ when the data error variances are known absolutely and used to compute the weights.[2] If the data error variances are known in only a relative sense, the parametric variances (from $\mathbf{V}_{\text{post}}$) become χ^2-distributed estimates that include the scale factor S/ν, with degrees of freedom ν equal to the difference in the numbers of data points and adjustable parameters, $\nu = n - p$. If the weights are not correct to within a constant, the parameter estimates remain normally distributed but are no longer minimum-variance or maximum-likelihood; and $\mathbf{V}$ is wrong (biased). The most common occurrence of this flaw is likely the use of $w_i = 1$ (unweighted LS) to fit data having nonconstant σ_{yi} (heteroscedastic data).

If the fit model includes nonlinear parameters, the fit becomes nonlinear (NLS) and the minimization of S generally requires iteration. Nonlinear estimators are typically biased and non-normal and often do not have finite variance. A simple example is the fit model, $y = 1/a$. Under the stated conditions, the reciprocal of a is the unbiased, minimum-variance estimate of the average $\bar{y}$, but a is biased with infinite variance.[3] Still, confidence limits can be established for a; and provided its SE is sufficiently small, the normal approximation can suffice for these. I have suggested a 10% guideline: If the parametric σ_a is $< |a|/10$, the 1-σ normal confidence limits are reliable to ∼10%.[4,5] These results still assume the data σ_{yi} and w_i are known absolutely; if not, the parametric SEs are estimates and the t-distribution is used for confidence limits.[1,6]

There are situations where the assumption of error-free independent variables holds rigorously, for example, in fitting spectroscopic data where the x_i are quantum numbers. On the other hand, in operations like calibration, the "knowns" have some uncertainty; and with the high-precision capabilities of

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many instruments, they can even become the dominant source of uncertainty, justifying reversing the regression.[7] A test for the successful use of eq 1 can be made by examining the relative magnitudes of the terms in the *effective variance*, which for two variables is

$$\mathrm{var}(y)_{\mathrm{eff}} = \sigma_y^{\ 2} + (\mathrm{d}y/\mathrm{d}x)^2\sigma_x^{\ 2} \tag{2}$$

If the second term is an order of magnitude smaller than the first for most data, the usual procedure should provide reliable results; if the reverse holds, regression of x on y should be considered.

There remain cases where the uncertainties are comparable in two or more variables, rendering the dependent/independent distinction meaningless and requiring more sophisticated treatment for reliable results. This case is the subject of the rest of this work, and I will start by asserting that the optimal solution is obtained by minimizing the extension of eq 1 to all uncertain variables,

$$S_{\mathrm{TV}} = \sum w_{xi}\delta_{xi}^{\ 2} + w_{yi}\delta_{yi}^{\ 2} + \ldots \tag{3}$$

where I use the subscript TV for "total variance", a label which has unfortunately been used in a more general sense at times in the statistical literature. The emphasis on S_{TV} as a minimization target dates back at least to Deming's work;[8] and general-purpose algorithms for accomplishing this have been available since the early 1970s, from Powell and McDonald[9] and Britt and Leucke.[10] Similar procedures were provided later by Southwell,[11] Jeffries,[12] Lybanon,[13] and Boggs et al.[14] The TV method has the satisfying property of yielding a single set of parameter estimates for any response function, linear or nonlinear, independent of the manner in which the relation or relations among the variables and parameters are expressed.[15] As before, optimal parameter estimates are assumed to require inverse-variance weighting, so pointwise variances are required for all uncertain variables; and again, all such variances must be known to within a single constant. Further, all uncertain variables must be "adjusted," meaning, e.g., an X_i for each x_i, giving residual $\delta_{xi} = X_i - x_i$, in addition to δ_{yi} given above.

This last aspect, the adjustment of all uncertain variables, makes these algorithms more complex than NLS algorithms for a single uncertain quantity. Accordingly some authors have pursued methods in which the uncertainties in the variables are cast into contributions to the variance in the quantity targeted for minimization, for which y is typically used, or the function itself in the case of implicit relations.[16−19] With y dependent and x the only other uncertain variable, the result is the effective variance given in eq 2. Taking the weights $w_{i,\mathrm{eff}} = 1/\mathrm{var}(y_i)_{\mathrm{eff}}$, the minimization target is

$$S_{\mathrm{EV}} = \sum w_{i,\mathrm{eff}}\delta_{yi}^{\ 2} \tag{4}$$

in which only the y residuals are needed, as in the usual LS treatments. The standard EV procedure involves (1) iterative minimization of S_{EV} with the $w_{i,\mathrm{eff}}$ treated as constant, followed by (2) adjustment of the $w_{i,\mathrm{eff}}$ to consistency with the new parameter estimates. This procedure is also known as iterative reweighting. Convergence typically requires 5−15 cycles.

The EV method is easy to use with standard programs for weighted LLS and NLS, but it has some known shortcomings:[20,21] (1) The minimum in S_{EV} is not the same as that in S_{TV}, and thus the solutions are not "optimal". (2) When the weights in x and y are constant (but not necessarily the same), the solution to the straight-line fit model becomes just the unweighted or ordinary LS (OLS) solution for regression of y upon x and converges in one cycle. This is true even when the uncertainties in x greatly exceed those in y, making regression of x upon y more sensible. (3) It follows that the results depend upon the manner in which the fit relation is expressed, in contrast with the unique results given by the TV method.

Through what may appear to be a small change in the EV procedures, we obtain results that are significantly different: Include the dependence of $w_{i,\mathrm{eff}}$ on the adjustable parameters in the optimization. For example, applying this method (EV_2) to the straight-line response function,

$$y = a + bx \tag{5}$$

makes the minimization target

$$S_{\mathrm{EV}_2} = \sum \frac{(a + bx_i - y_i)^2}{\sigma_{yi}^{\ 2} + b^2\sigma_{xi}^{\ 2}} \tag{6}$$

Note that no consistency iteration is required now, since the b-dependence in $w_{i,\mathrm{eff}}$ is included in the minimization. In Monte Carlo (MC) comparisons of the TV and EV_2 methods on a number of nonlinear models, I concluded that the differences were unlikely to be significant in practical applications, where uncertainties from estimating the data variances would usually be greater.[15] Further, as I have illustrated for the KaleidaGraph and Excel programs,[22] the EV_2 method is easy to use for any response function and should arguably become the default method for treating problems with multiple uncertain variables.

Importantly, the EV_2 and TV solutions are identical for straight-line response functions. This holds for *any* straight-line response function. For example, taking x residuals as $x - y/b + a/b$ and redefining the $w_{i,\mathrm{eff}}$ accordingly gives results identical to those from y residuals in eq 6. Williamson is often cited by physical scientists for showing this TV/EV_2 equivalence,[23] in his treatment of York's widely used weights[24] for a model originally posed by Pearson.[25] Sprent called this GLS ("generalized least squares"),[26] but that term has also been used for a procedure designed for simultaneous estimation of data variance functions and parameters.[27] Lisý et al. called it BLS ("bivariate least squares").[28] In fact, Deming gave the TV/EV_2 equivalence for a linear response function as his Exercise 3 in Section 58,[8] and he cited an 1879 paper by Kummel for it.[29]

To recap briefly to this point, the TV and EV methods apply for any response function and any number of uncertain variables. The GLS and BLS methods are similar but are described for straight-line response functions only, for which they give solutions identical to those from TV and EV_2. All of these require pointwise information about the variances in x and y, and they are implemented with iterative numerical procedures. On the other hand, it *is* possible to obtain closed-form solutions for the straight-line response function by making simplifying assumptions about the data errors. The resulting methods are often used without regard for these assumptions, so I will call them "ignorance methods."

The CVR (constant variance ratio) method requires the ratio $\lambda = \sigma_y^{\ 2}/\sigma_x^{\ 2}$, which is assumed to be the same for all data. Under these conditions, it gives parameter estimates that agree exactly with those from the TV, EV_2, GLS, and BLS methods, for which I will henceforth just use TV as a reference. In fact, although this variance ratio is generally considered to apply for constant σ_x and σ_y, the agreement holds also for heteroscedastic σ_x and σ_y, provided all (x_i,y_i) points have the same λ and EV weighting is used (see below). Orthogonal regression (OR) targets the

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Table 1. Least-Squares Methods Considered

	Method	Key properties/conditions
LLS	Linear least squares	Single uncertain variable (y) and response function algebraically linear in adjustable parameters; includes weighted LS (WLS, variable w_i) and ordinary LS (OLS, $w_i = 1$).
NLS	Nonlinear least squares	Response function nonlinear in adjustable parameters; requires iteration from initial parameter estimates; handles any weighting; usually assumes single uncertain variable.
TV	Total variance method	NLS with multiple uncertain variables and any response function. Minimizes S_{TV} (eq 3), for which pointwise estimates of data error (σ_{xi}, σ_{yi}, ...) are required. Yields unique solution for parameters and their SEs, for any way of expressing the response function.
EV	Effective variance method	Like TV but treated as a single uncertain variable or function, with uncertainties from all other uncertain variables projected into that for designated quantity, giving weights $w_{i,\mathrm{eff}} \propto$ inverse effective variance. The $w_{i,\mathrm{eff}}$ must be iterated to consistency with the parameter values.
EV_2	Modified EV method	Like EV, but $w_{i,\mathrm{eff}}$ included directly in minimization. Solutions for linear response functions identical to those from TV method.
GLS	Generalized least squares	Defined for linear response function of two variables. Minimization target and solutions identical to those from EV_2 method.
BLS	Bivariate least squares	GLS.
CVR	Constant variance ratio	Like GLS/BLS but assumes constant ratio $\lambda = \sigma_y^2/\sigma_x^2$, for which it agrees exactly with TV/EV_2/GLS/BLS. Although derived for constant σ_y and σ_x, this agreement holds also for variable σ_{xi} and σ_{yi}, provided their ratios are the same for all i and EV weighting is used.
DR	Deming regression	CVR.
OR	Orthogonal regression	CVR with $\lambda = 1$.
TR	Triangular regression	Linear response function; minimizes sum of triangular areas between points and fitted line. Agrees with CVR under special limiting conditions, also for variable σ_{xi} and σ_{yi} having constant λ, under EV weighting.
GMR	Geometric mean regression	TR. So named because slope is geometric mean of OLS slopes for regression of y on x and x on y. This relation holds also for weighted data, provided the WLS weights for the two fits are proportional.
MAR	Major axis regression	TR.
MM	Method of moments	Linear response and known σ_x, giving parameters and σ_y. Parameters agree with TV/CVR under special conditions, but SEs do not.

squared orthogonal distances for minimization and agrees with CVR and hence TV when $\lambda = 1$. Triangular regression (TR), also called geometric mean regression (GMR) and major axis regression (MAR), minimizes the sum of the triangular areas between the points and the fitted line. It agrees with CVR (hence TV) when the error variance ratio λ equals the population variance ratio S_{yy}/S_{xx} (see below); again this holds also for heteroscedastic data under EV weighting. The method of moments (MM) requires the variance for x (assumed constant) and estimates σ_y^2, for which value it agrees with TV; accordingly it does not agree for other σ_y. However, as with the CVR method, it does agree with TV for heteroscedastic data when the σ_{yi} for the TV calculation are set to be consistent with the MM results and EV weighting is used.

All of these methods have long histories, but unfortunately the labeling has not been used consistently. The CVR and OR methods date back to Kummel,[29] Adcock,[30] and Pearson.[25] TR appears to have been developed independently by a number of authors,[31] the earliest perhaps being Sverdrup.[32] The CVR method is called MM by Fuller (section 1.3),[33] who cites Koopmans[34] for the earliest detailed treatment. Carroll and Rupert[35] call it OR, while "weighted orthogonal distance regression" (ODR)[14] is equivalent to TV (and is similarly useable on nonlinear response functions). The equations generally used for implementing the MM method are those given in section 1.2.2 and example 1.2.1 by Fuller.[33,36,37] "Deming regression" (DR) is often taken to mean results given by Deming[8] for straight-line fitting in his exercise 6 in section 65, which in fact are CVR results. Deming's development is much more general, intended for any response function and any number of uncertain variables; and apart from a small correction,[15] it agrees with TV. For linear response functions, his general treatment is equivalent to TV.

Although I have referred to OR, TR, CVR, and MM as ignorance methods, they *can* be implemented with weighting, by just including a factor of w_i within each evaluated sum (see below). Also, if the EV weights $w_{i,\mathrm{eff}}$ are used, these methods give results that often compare well statistically with TV results, as found through MC simulations to be described below. Indeed the agreement can be exact under special forms of heteroscedasticity, as noted earlier. Of course, EV weighting requires information about the σ_{xi} and σ_{yi}, negating the "ignorance" label. Since the EV_2 method is so easy to use, there is little practical need for this result; still, this improved performance under EV weighting is noteworthy, intellectually satisfying, and apparently not previously recognized.

The methods under comparison here are summarized in Table 1. In subsequent sections, I review the working relations of these methods and then compare their performance on straight-line fit models, including the York-weights version[24] of Pearson's example,[25] which has served as a touchstone in many studies of methods for dealing with error in x and y. I will emphasize estimation of both the parameters and their SEs. In the course of this work, I have checked a number of numerical examples from the literature; and I have found some disagreements in estimates of particularly the SEs. I will include a tally of these.

■ COMPUTATIONAL RELATIONS

In the following, I will neglect two concerns often included in statistical treatments, correlation between the errors in x_i and y_i and equation error. In many situations of interest in scientific data analysis, x and y are measured separately and independently, so that no correlation in their measurement errors is expected. If such correlation can in fact be discerned and determined, it can be accommodated in the EV_2 minimization target of eq 6, by adding the term $-2b\sigma_{xyi}^2$ to

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the denominator, where σ_{xyi}^2 is the covariance of the errors in x_i and y_i.[37] In principle, equation error can be included in the total variance estimates through measurements that replicate all aspects of the experiment. Because all methods except TV and EV apply only for the linear response function, I will confine my attention to the model of eq 5, using a and b for estimated values, with subscript t added for "true," where needed.

Linear Least-Squares. Minimization of S in eq 1 with respect to a and b gives the equations,

$$a \sum w_i + b \sum w_i x_i = \sum w_i y_i \tag{7a}$$

$$a \sum w_i x_i + b \sum w_i x_i^2 = \sum w_i x_i y_i \tag{7b}$$

which can be written in matrix form as

$$\begin{pmatrix} \sum w_i & \sum w_i x_i \\ \sum w_i x_i & \sum w_i x_i^2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \sum w_i y_i \\ \sum w_i x_i y_i \end{pmatrix} \equiv \mathbf{A}\boldsymbol{\beta} = \mathbf{z} \tag{8}$$

the solution of which is $\boldsymbol{\beta} = \mathbf{A}^{-1}\,\mathbf{z}$, where $\mathbf{A}^{-1}$ is the matrix inverse of $\mathbf{A}$. If the data variances are known and the weights are taken as their inverses, $w_i = \sigma_{yi}^{-2}$, the covariance matrix

$$\mathbf{V}_{\text{prior}} = \mathbf{A}^{-1} \tag{9}$$

is exact, with diagonal elements σ_a^2 and σ_b^2. If the data variances are known relatively but not absolutely (as from a data transformation, e.g., y to $\ln(y)$), the post estimates, s_a^2 and s_b^2, are the diagonal elements of $\mathbf{V}_{\text{post}}$, obtained by scaling $\mathbf{A}^{-1}$ by s_y^2 from

$$s_y^2 = \frac{\sum w_i \delta_i^2}{n - p} \tag{10}$$

Then s_y^2 is known as the estimated variance for data of weight unity; for inverse-variance weighting with known data variances, it is an estimate of the reduced χ^2, having average 1 and standard deviation (SD) $(2/\nu)^{1/2}$. As long as the w_i are proportional to σ_{yi}^{-2}, $\mathbf{V}_{\text{post}}$ (for *a posteriori*) is an uncertain estimate of $\mathbf{V}_{\text{prior}}$ (*a priori*); with any other weighting $\mathbf{V}_{\text{post}}$ is biased. OLS (ordinary LS) is a special case of weighted LS, with $w_i = 1$, and is valid only if the data are truly homoscedastic. The consequences of weighting error range from nominal to very significant.[38]

Confidence limits are defined to encompass a specified percentage of possible outcomes (usually 90% or 95%) and are obtained from the normal (Gaussian) distribution,

$$P_G(\mu, \sigma, x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \tag{11}$$

when $\mathbf{V}_{\text{prior}}$ applies, where μ is the mean and σ the SD, or for $\mathbf{V}_{\text{post}}$, the t-distribution,

$$P_t(t) = C(1 + t^2/\nu)^{-(\nu+1)/2} \tag{12}$$

where C is a normalizing constant, $t = (\beta - \beta_t)/s_\beta$, and the number of degrees of statistical freedom $\nu = n - 2$ here. The t-distribution converges on the normal distribution in the limit $\nu = \infty$.

Variance estimates (s_y^2) follow a scaled χ^2 distribution,

$$P_{\text{rcs}}(z) = C' z^{(\nu-2)/2} \exp(-\nu z/2) \tag{13}$$

where C' is another normalization constant and $z = (\text{var})/\alpha$, with α a scaling factor (equal to the mean of (var)). $P_{\text{rcs}}(z)$ also becomes normal in the large-ν limit.

Technically, normal data error is required to give parameter distributions that follow eqs 11 and 12 and χ^2-distributed s_y^2. Concern about non-normal data is probably overblown, since nowadays data are often collected by instruments that average multiple readings; and even if not, the parameter distributions become normal as ν increases to $\gtrsim$25. Both of these results follow from the central limit theorem.[5,6] However, normality is not needed for LLS to return unbiased, minimum-variance parameter estimates.

The literature expressions for the parameter estimates and SEs in the ignorance methods are given in terms of sums like those that appear in $\mathbf{A}$ and $\mathbf{z}$, so it is useful to give those results here:

$$b = S_{xy}/S_{xx} \tag{14a}$$

$$a = \bar{y} - b\bar{x} \tag{14b}$$

with $\bar{x} = \sum w_i x_i / \sum w_i$ and similarly for $\bar{y}$, $S_{xx} = \sum w_i (x_i - \bar{x})^2$ and similarly for S_{yy}, and $S_{xy} = \sum w_i (x_i - \bar{x})(y_i - \bar{y})$. The prior variances are $\sigma_a^2 = A_{22}/|\mathbf{A}|$ and $\sigma_b^2 = A_{11}/|\mathbf{A}|$, with $|\mathbf{A}| = A_{11}A_{22} - A_{12}^2 = S_{xx}\,S_w$, with $S_w = \Sigma w_i$. Note that these variances do not depend on the y_i. The matrices $\mathbf{A}$ and $\mathbf{z}$ can themselves be generated from matrices containing the data.[2]

Nonlinear Least-Squares. The TV, EV, EV$_2$, GLS, and BLS methods are nonlinear, so they require iteration. The mathematical operations involve matrices analogous to $\mathbf{A}$ and $\mathbf{z}$; and at convergence, they have the same significance, i.e., $\mathbf{V}_{\text{prior}} = \mathbf{A}^{-1}$ and $\mathbf{V}_{\text{post}} = s_y^2\,\mathbf{A}^{-1}$.[4,5] In particular, this holds for multiple uncertain variables in the TV method, where for $\mathbf{V}_{\text{post}}$, s_y^2 is replaced by S_{TV}/ν.[15] As already noted, NLS parameter estimates are inherently non-normal in distribution but may be approximated as normal when the SEs are sufficiently small, $^1/_{10}$ the magnitude of the parameter being a useful guideline.[5] The χ^2 distribution typically better approximates S_{TV} estimates than the normal distribution does for parameters.[15]

CVR, OR, TR, and MM. The CVR method assumes constant σ_y^2 and σ_x^2. Mandel[39] gave a thorough treatment of this method, including correlated error in x_i and y_i. Setting this correlation to zero, the slope b is the solution of a quadratic equation,

$$b_{\text{CVR}} = \frac{1}{2S_{xy}}\{B + [B^2 + 4\lambda S_{xy}^2]^{1/2}\} \tag{15}$$

where $\lambda = \sigma_y^2/\sigma_x^2$ and $B = S_{yy} - \lambda S_{xx}$. a_{CVR} is then given by eq 14b. These results agree with Deming's in exercise 6 of section 65,[8] with Deming's c being the reciprocal of λ.

Mandel's expressions for the SEs require quantities S_{uu} and S_{vv}, given by

$$S_{uu} = S_{xx} + 2bS_{xy}/\lambda + (b/\lambda)^2 S_{yy} \tag{16a}$$

and

$$S_{vv} = b^2 S_{xx} - 2bS_{xy} + S_{yy} \tag{16b}$$

from which s_e is defined as $s_e = (S_{vv}/\nu)^{1/2}$. The parameter SEs are

$$s_{a,\text{CVR}} = s_e\left(\frac{1}{S_w} + \frac{\bar{x}^2(1 + b_{\text{CVR}}^2/\lambda)^2}{S_{uu}}\right)^{1/2} \tag{17a}$$

and

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$$s_{b,\mathrm{CVR}} = s_e \frac{1 + b_{\mathrm{CVR}}{}^2/\lambda}{\sqrt{S_{uu}}} \tag{17b}$$

As already noted, both homoscedastic and heteroscedastic data having $\sigma_{yi} = \lambda\sigma_{xi}$ give exact agreement with TV for both the parameters and their SEs when weighted using $w_i = w_{i,\mathrm{eff}}$. Then S_{vv} from eq 16b equals the sum in eq 10, so $s_e = \sigma_y$. Accordingly, the SEs in eqs 17 are post-SEs, and the prior SEs are obtained by omitting s_e.

Since OR is just CVR with $\lambda = 1$, eqs 15−17 hold also for OR.

For the TR method, the GMR label follows from the property that the estimated slope is the geometric mean of those for OLS regression of y on x and x on y.[31,40,41] Or if the data are displayed with reversed axes in the latter case, this is $b_{\mathrm{TR}} = \pm(b_{y/x}/b_{x/y})^{1/2}$, or

$$b_{\mathrm{TR}} = \mathrm{sign}(S_{xy})(S_{yy}/S_{xx})^{1/2} \tag{18}$$

with a_{TR} again given by eq 14b. Although these relations were derived for homoscedastic data, eq 18 remains valid for heteroscedastic data. Also, the geometric mean relationship continues to hold for weighted data, provided the weights for the two WLS fits are proportional. Further, if the error variance ratio $\lambda = S_{yy}/S_{xx}$, eq 15 for b_{CVR} reduces to eq 18 for b_{TR}.

Patzer et al.[31] have given expressions for the TR parameter variances,

$$s_{b,\mathrm{TR}}{}^2 = \frac{2b_{\mathrm{TR}}}{\nu}(b_{\mathrm{TR}} - S_{xy}/S_{xx}) \tag{19a}$$

$$s_{a,\mathrm{TR}}{}^2 = s_{b,\mathrm{TR}}{}^2\left(\frac{S_{xx}}{S_w} + \bar{x}^2\right) \tag{19b}$$

where the latter expression has S_w in the denominator rather than n, to accommodate weighting, and where the factor of 2 in the numerator of eq 19a corrects for an apparent factor-of-2 mistake in Table 2 of ref 31. These are post-SEs. For the purpose of MC simulations, we need prior-SEs in terms of σ_{xi} and σ_{yi}. By propagating the errors in x_i and y_i into a and b, I obtain

$$\sigma_{b,\mathrm{TR}}{}^2 = \frac{1}{S_{xx}S_{yy}} \sum w_i{}^2[\sigma_{yi}{}^2(y_i - \bar{y})^2 + b_{\mathrm{TR}}{}^4\sigma_{xi}{}^2(x_i - \bar{x})^2] \tag{20a}$$

$$\sigma_{a,\mathrm{TR}}{}^2 = \bar{x}^2\sigma_{b,\mathrm{TR}}{}^2 + 1/S_w \tag{20b}$$

where the latter expression holds rigorously for exactly fitting data and is a very close approximation for actual data (from numerical tests). The scale factor relating the prior and post parameter variances is $\Sigma w_i A_i/\nu$, where A_i is the area of the triangle formed by the ith point and the fitted line; this is again an approximation, good to ∼1%.

In the method of moments, like the other methods in this section, the x and y data are assumed to be homoscedastic. Modifying the expressions from Fuller (section 1.2.1) for weighting, I obtain

$$b_{\mathrm{MM}} = \frac{S_{xy}{}'}{S_{xx}{}' - \overline{\sigma_x^2}} \tag{21}$$

with a_{MM} again given by eq 14b. Here the primes designate sums scaled by the factor $\frac{n}{(n-1)S_w}$, and the average is a simple average of the x variances. The y data variance is estimated from

$$s_y{}^2 = S_{yy}{}' - b^2(S_{xx}{}' - \overline{\sigma_x^2}) \tag{22}$$

and the parameter SEs from

$$s_{b,\mathrm{MM}}{}^2 = \frac{S_{xx}{}'S_{vv}{}' + (b\overline{\sigma_x^2})^2}{(n-1)(S_{xx}{}' - \overline{\sigma_x^2})^2} \tag{23a}$$

and

$$s_{a,\mathrm{MM}}{}^2 = s_{b,\mathrm{MM}}{}^2\bar{x}^2 + S_{vv}{}'/n \tag{23b}$$

where

$$S_{vv}{}' = \frac{n-1}{n-2}(S_{yy}{}' + b^2S_{xx}{}' - 2bS_{xy}{}') \tag{24}$$

As already noted, the MM parameters agree with TV estimates when, for the latter the σ_y values are set to those from eq 22 for a given σ_x. As for the CVR method, this situation continues to hold for heteroscedastic data under EV weighting (see below). However, the parameter SEs do not agree in either situation.

It is noteworthy that for any heteroscedastic error structure for x and y, the prior SEs are identical for the TV, CVR, OR, and TR methods for exactly fitting data, when EV weighting is used for the last 3. This does not hold for the MM method, because this method does not fit exact data, as eq 21 forces the slope to be smaller than the true slope unless $\sigma_x = 0$.

Algorithms. All computations for the four methods discussed in the preceding section were done by coding the expressions in eqs 15−24. Most of the results were obtained using EV weighting, in the iterative reweighting mode of the EV method of eq 4. For the CVR, OR, and TR methods, the weights drop out of the parameter calculations for homoscedastic σ_x and σ_y, just as they do in the EV method. (However, they do affect the SEs, giving reliable results.) To check on the possibility that including the w_i directly in the minimizations might improve results (as in going from eq 4 to its EV_2 version in eq 6), I used (with modifications) the FRPRMN routine from Press et al.[42] to minimize the sum of weighted, squared orthogonal distances (for OR) and triangular areas (for TR). After confirming that this method yielded correct EV_2 results when the sum in eq 6 was the minimization target, I found no clear improvement and even some worsening in the MC results for TR and OR, so I did not pursue this approach further.

The TV/EV_2 MC calculations were done with a FORTRAN algorithm like that described for EV_2 with KaleidaGraph and Excel in ref 22. A FORTRAN TV algorithm was used at times to double check results and for response functions other than the straight line.

Monte Carlo simulations were done by adding random Gaussian error of magnitudes σ_{xi} and σ_{yi} to true values $x_{t,i}$ and $y_{t,i}$, then computing the parameters and SEs as, e.g., in ref 15. Results were accumulated, statistically assessed, and examined for bias in the parameters and their SEs, and for normality of the parameters and the χ^2 distribution for S_{TV}. Computations were done for (a) a 5-point linear model having an intercept of 1, varying slopes, and varying errors in x and y and (b) for the 10-point Pearson/York model.[24,25] Most runs comprised 4×10^4 synthetic data sets.

RESULTS AND DISCUSSION

5-Point Linear Model. Figure 1 shows biases in the slope b and its SD, from MC simulations on the 5-point linear model for a range of constant σ_x,σ_y values and slopes, for three of the methods. (Here CVR is identical to TV in every case, and OR =

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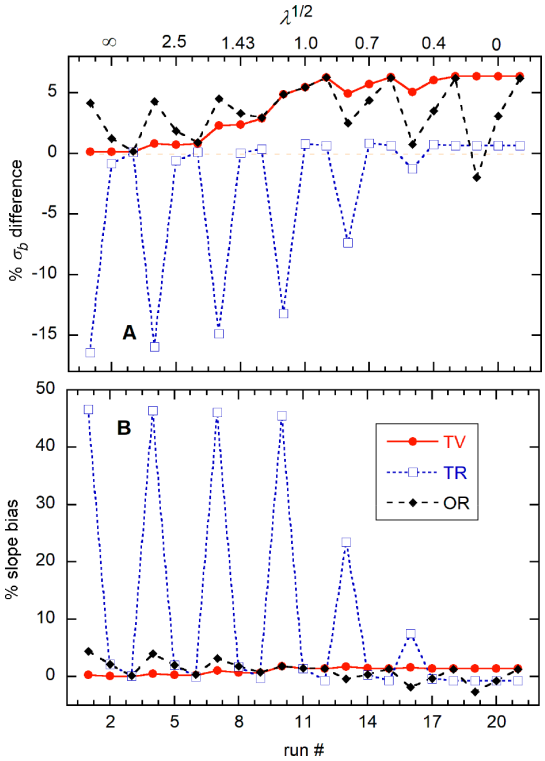

Figure 1. Biases in the slope and its SD from MC simulations for a 5-point linear model having true x_i = 1.1, 3.3, 5.5, 8.3, and 12 and intercept 1.0. The references are the prior σ_b and b_t; e.g., % slope bias = $(\bar{b}_{MC}/b_t - 1) \times 100\%$. In each set of 3 runs, the slopes are b_t = 0.2, 1.0, and 5.0, respectively. For sets 1−3, 4−6, 7−9, etc., the data errors are (σ_x,σ_y) = (0,1), (0.4,1), (0.7,1), (1,1), (1,0.7), (1,0.4), and (1,0), respectively (ratios at top). The MC statistical uncertainties are comparable to the size of the displayed points.

TV for the middle points, where $\sigma_x = \sigma_y$.) Interestingly, TV is not always "best," for example, showing greater bias than TR and OR in some cases and lower precision than TR in all cases. The latter observation is particularly interesting for the first 3 points, where TV becomes OLS, which is said to give unbiased, minimum-variance estimates. The SD for TR for the first point is 17% smaller; however, the total variance, taken as the sum, bias$^2 + s_b^2$, does exceed that for TV. At the other extreme (σ_y = 0), TV becomes OLS for regression of x on y, which means that the slope b is the reciprocal of the normal quantity $1/b$. For all b with σ_x = 1 and σ_y = 0, the TV slopes have predicted 12% uncertainty, which is enough to give the observed biases in b_{TV} and its SD.[3]

The parameter distributions for b_{TV} and b_{TR} from runs 1 and 19 are illustrated in Figure 2. Only $b_{TV,1}$ is normal, but $b_{TV,19}$ does follow the reciprocal normal distribution,[3,4] as expected (see the Supporting Information). Interestingly, the OR b estimates are also normal for σ_x = 0 and reciprocal normal for σ_y = 0, though biased for both. The χ^2 distribution holds for S_{TV} in both cases and is a close approximation for $S_{\nu\nu}$ from OR (Supporting Information). These examples are considered further in the Supporting Information for more extreme behavior when the data error becomes much larger. The Supporting Information also includes a different presentation of the information in Figure 1, as 9 curves for the 9 combinations of method and slope, displayed as functions of $\lambda^{1/2} = \sigma_y/\sigma_x$. Note that the effects

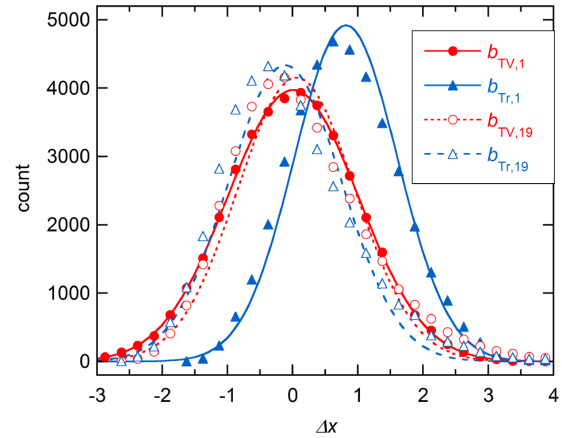

Figure 2. Histogrammed slopes from Monte Carlo simulations from runs 1 and 19 in Figure 1, with normal fits obtained using Poisson weighting (w_i = inverse count). The Δx axis is relative to the true value of the slope (0.2), in units of the predicted SEs (0.1172 and 0.02344). The χ^2 value for the Gaussian fit for $b_{TV,1}$ is 27.3, which is close to the expected value for 32 points and 3 parameters, supporting normality. The other 3 fits give χ^2 values >1000, confirming their non-normality.

illustrated in Figure 1, though highly significant for 40 K simulations, are mostly of little practical significance. Thus, even the 46% bias in b_{TR} from run 1 exceeds b_t by less than 1σ for a single data set.

Results for the same 5-point model with a heteroscedastic data error are shown in Figure 3. The TR and OR calculations

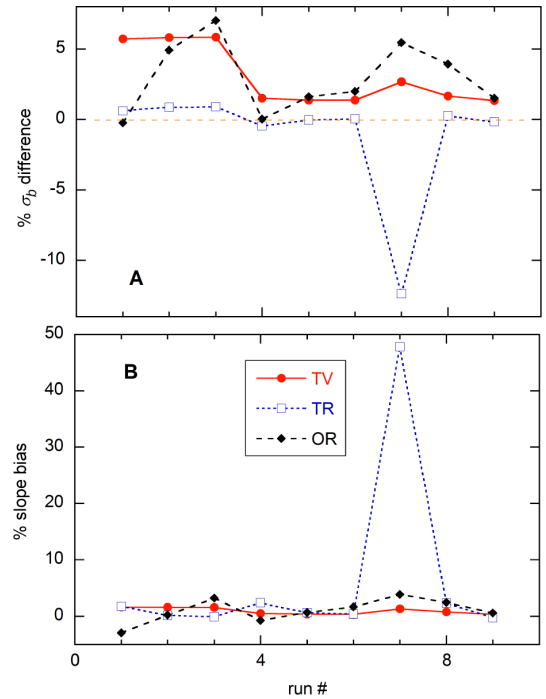

Figure 3. Monte Carlo simulation results for slope in 5-point linear model of Figure 1, having true slopes of 0.2, 1.0, and 5.0, in each set of 3 results. For sets 1−3, 4−6, and 7−9, the data errors are (σ_x,σ_y) = (1,0.1 × y_t), (0.1 × x_t, 0.1 × y_t), and (0.1 × x_t,1), respectively.

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employed EV weighting, without which the results are much poorer. For example, when run 5 (10% error in both x and y, b_t = 1) was repeated with neglect of weighting, there was ∼50% increase in the SD for b for both the TR and OR methods and 45% for TV. The outstanding anomaly in Figure 3 is again in the TR results for b_t = 0.2, but it manifests only in the 10% σ_x results here. As before, the bias more than offsets the precision gain.

A few of the runs in Figure 1 were repeated for the MM method, with results generally slightly worse than OR on bias and precision loss but with none of the anomalies that occur for TR. For σ_x = 0, the parameters are those for OLS, so the MC results agree exactly with TV. The average estimated σ_y^2 was always low, 0.75, for example, in the OLS example with input σ_y = 1. This appears to be a degrees of freedom issue, with $\nu = n - 1$ = 4 being the factor in Fuller's equations for the primed sums in eqs 21−24, while ν = 3 is the correct value to estimate σ_y^2 from the sum of squared residuals in eq 10. The SEs from eqs 23 did reliably predict the MC SEs (within 1%).

The most striking results from Figures 1 and 3 are the strong biases for TR when b_t = 0.2. This behavior was missed by Francq and Govaerts, who presented results from similar computations, but just for a_t = 0 and b_t = 1, in keeping with their emphasis on methods comparisons.[43] In checking their model for b_t = 0.2 at λ = 20 and n = 10, I obtained average (a,b) estimates of (−0.004,0.20), (−3.8,0.46), and (−0.50,0.23), respectively for TV, TR, and OR. Both TR biases exceed $2\sigma_{a,b}$ in magnitude, meaning this method simply fails in this case. Francq and Govaerts also investigated the dependence on the number of fitted points (n = 10, 20, 50), confirming that the biases *increase* with n for the TR and OR methods, meaning these estimators are *inconsistent*. (The TV method gives consistent parameter estimates, i.e., the biases vanish in the large-n limit.)

York Model. The 10-point York/Pearson problem is heteroscedastic in both x and y, with w_x decreasing from 1000 to 1 while w_y increases from 1 to 500.[24,25] Thus, none of the ignorance methods can give exact agreement with TV. MC simulated data were analyzed with all 5 methods, using the model from ref 15: $a_t = 5.5$, $b_t = -0.48$, $y_{ti} = a_t + b_t x_i$, $\sigma_{xi} = w_{xi}^{-1/2}$, and $\sigma_{yi} = w_{yi}^{-1/2}$. The biases in parameters and SEs were at most 0.3% for the TV method. For the others, the slope biases ranged from −5.6% for MM to +1.0% for CVR, while the precision losses were 2.1% (TR) to 14% (MM). With unweighted analysis, the biases changed little, but the precision losses were 49% (TR) to 55% (MM and OR).

Results for the actual York data are given in the Supporting InformationTable S-1 , with sufficient computational precision to match results from refs 44 and 45 for the TV method. With EV weighting, all other methods give results close enough to TV to be considered reasonable.

Literature Critique. There are very many published papers dealing with the problem of estimating parameters when multiple variables are uncertain, and I have cited what I think are key contributions. Here I want to distinguish between methods and algorithms. Thus, I consider any algorithm that gives numerical results for the York problem that agree with the TV results as employing either the TV method or its equivalent for linear response functions, the EV_2, GLS, and BLS methods. Indeed, as I have noted, one can obtain these parameters through brute-force minimization of the target function S_{EV_2} of eq 6. Thus, all but 5 of the 14 techniques compared in Table 2 of ref 44 employ either TV or EV_2, and the 5 (including OLS, WLS, and EV) are clearly not minimizing S_{EV_2}. Including the SEs in the comparisons can reveal further differences. For example, in ref 2 of ref 44, Kalantar[46] obtained TV parameters but not the **V**-based SEs, as I have verified with my own computations; and Kalantar cited Williamson for his SE expressions. Similarly, by obtaining 6−8 digit agreement with the parameters and SEs in Appendix C of Boggs et al.[14] for the 3-parameter exponential model of their Table I (from Draper and Smith[47]), I can confirm that their ODRPACK produces TV results. This means that it should also give identical results with the response function expressed in logarithmic form, though I have not attempted to check the ODRPACK program for that.

Using literature data for the determination of arsenic in water by two methods, Ripley and Thompson[48] analyzed the data (30 points heteroscedastic in both variables) by OLS, WLS, and IWLS (iteratively weighted LS), for both y/x and x/y regression in each case and by MLFR (maximum likelihood fitting of a functional relationship). I have confirmed their results in all cases and shown that IWLS = EV and MLFR = EV_2. They reported SEs only for MLFR, and these are the prior values. The same data have been used recently as an example in an R program "deming."[49] The parameters agree with those from MLFR, confirming that this is a TV/EV_2 method, not the more limited CVR method many attribute to Deming. However, the SEs and confidence limits in the output appear to be wrong.

Fuller's example 1.2.1[33] employs the method now commonly known as MM. σ_x^2 is assumed known, and the analysis output includes an estimate of σ_y^2. If this value is taken as known in an EV_2 analysis, the resulting parameter estimates agree with those from MM, but the SEs do not. Thus, the EV_2 post SEs are s_a = 10.31 and s_b = 0.1422, while Fuller's estimates are larger by 22%. Of course, any other assumed σ_y will give different EV_2 parameters and SEs. Fuller's example 1.3.1 assumes the data variance ratio is known, now commonly called the CVR method; and indeed both EV_2 and CVR give his results for his assumed λ. However, again the SEs differ: The (post) estimates from EV_2 and CVR agree exactly but Fuller's exceed them by 4.0%.

Examples of heteroscedastic x and y data error are relatively rare in the literature, notable exceptions being the York model and the Ripley and Thompson example. Another is by Cheng and Riu,[37] who used 6 methods in a methods comparison of two sets of results for analysis of Ca^{2+} in water and reported the results in their Table 4. I have confirmed their parameter results for GLS (EV_2), MM, CVR, and OR, which methods were used unweighted, with the average of the σ_y^2 and σ_x^2 values used to compute λ in the CVR calculation. However, I have failed to reproduce their SEs in any of these cases. Also, their ODR results should be identical to GLS, since they used the ODRPACK program,[14] which is TV. I have given complete results for these methods and TR in the Supporting Information Table S-1, for both weighted and unweighted analysis by the TR, CVR, OR, and MM methods. These results further confirm that the ignorance methods agree much better with TV when EV weighting is used.

Weighting *has* been used previously for at least one of the ignorance methods. Linnet treated proportional data error in both x and y by the CVR method, taking both data errors as proportional to $(x_i + y_i)$, with the proportionality constants determining λ.[51,52] The application was again methods comparison, so he used the model (a_t,b_t) = (0,1). For this model, MC simulations give identical results for Linnet's weighting and EV weighting. However, the agreement vanishes for MC simulations with $a_t \neq 0$ or for any "real" data set. Of

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course this weighting is suitable only for proportional error in both variables, while EV weighting is general.

CONCLUSION

In a critical comparison of the performance of several methods for estimating parameters and their SEs in the fitting of data having uncertainty in both x and y to a straight line, the CVR, OR, and MM methods compare favorably with the EV_2 method for homoscedastic error in the variables, the TR method less so because of larger bias under some conditions. In their usual implementation, all of the "ignorance" methods show significant precision loss when one or both variables have heteroscedastic error. However, if these methods are used with EV weighting, their performance again becomes comparable to that of the EV_2 method and, under some circumstances, exactly the same. These results seem not to have been noted previously.

The use of EV weighting requires knowledge of σ_{xi} and σ_{yi} for all data, information that is always required for the TV, EV, and GLS/BLS methods. This need for information about σ_{xi} and σ_{yi} may seem an unpleasant inconvenience, but it should be a standard part of every data analysis; at the least, the analyst should determine whether the data are approximately homoscedastic or clearly heteroscedastic. The latter situation arises any time initially homoscedastic data are transformed for analysis, *e.g.*, by reciprocation or logarithmic transformation.[3−5] Knowledge of all data variances to within a single scaling constant is required for minimum-variance parameter estimation. However, high precision is not required here, as near-minimum-variance results can be obtained with only approximate information about the data error variances.[38,50]

The most common use of x-and-y error fitting in analytical chemistry has probably been in methods comparison studies, often done with simplifying assumptions about the uncertainties in x and y.[43,48,51−53] There are also calibration applications with non-negligible error in x.[44] With the high precision capabilities of modern instrumentation, this may be ever more the case, calling for methods like those discussed here, or when y is much more precise than x, just reversing the regression.[7]

Although the EV_2 method is not always "best" in the comparisons made here, it might be considered "safest," as its bias and precision loss is modest, with no clear dependence on the parameters and predictable dependence on the data error variance. It is also easy to code for computation in Excel and in programs that permit user-defined response functions, like KaleidaGraph.[22] Further, it is not limited to the straight-line response function, and its performance with nonlinear response functions is comparable to that for TV.[15] Nor is it limited to just two variables. I am unaware of any performance testing or TV/EV comparisons for 3 or more uncertain variables. However, a quick check of a simple bilinear model, $y = a + bx1 + cx2$, with y, $x1$, and $x2$ all uncertain, confirms that TV and EV_2 give identical results here, just as for the simpler 2-variable model. Thus, it seems likely that EV_2 can be used successfully on more complex multivariate models with multiple uncertain variables. Users analyzing data having comparable uncertainty in multiple variables should give it serious consideration.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.analchem.0c02178.

Illustration and discussion of the normal and non-normal parameter distributions from the MC simulations and of the χ^2 distribution for S_{TV} and S_{vv} (OR); complete results for analysis of the York model and Data set 2 from ref 37 by the TV, TR, CVR, OR, and MM methods, with and without EV weighting for all but TV (PDF)

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Notes

The author declares no competing financial interest.

Biography

Joel Tellinghuisen is a Professor of Chemistry, Emeritus at Vanderbilt University. He cut his data analysis teeth on problems in optical spectroscopy. More recently, he has emphasized problems in other areas, including isothermal titration calorimetry (ITC) and analytical applications of polymerase chain reaction (PCR).

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